

E.G.S.PILLAY ENGINEERING COLLEGE (Autonomous)

Approved by AICTE, New Delhi| Affiliated to Anna University, Chennai Accredited by
NAAC with „A Grade| Accredited by NBA (CSE, ECE, IT)

NAGAPATTINAM – 611 002



B.E. Biomedical Engineering

Second Year – Fourth Semester

Course Code	Course Name	L	T	P	C	Maximum Marks		
						CA	ES	Total
Theory Course								
1901MA401	Probability and Stochastic Processes	3	2	0	4	40	60	100
1902BM401	Bio Mechanics	3	2	0	3	40	60	100
1902BM402	Basics of Pathology and Microbiology	3	0	0	3	40	60	100
1902BM403	Control Systems	3	0	0	3	40	60	100
1902BM404	Biomedical Instrumentation	3	0	0	3	40	60	100
1902BM405	Digital Electronics and Integrated Circuits	3	0	0	3	40	60	100
Laboratory Course								
1902BM451	Pathology and Microbiology Laboratory	0	0	2	1	50	50	100
1902BM452	Biomedical Instrumentation Laboratory	0	0	2	1	50	50	100
1902BM453	Analog and Digital and Integrated Circuits Laboratory	0	0	2	1	50	50	100
1902GE451	Life Skills: Verbal Reasoning	0	0	2	1	100	-	100
Mandatory Course								
1901MCX01	Environmental Science	2	0	0	0			
	Total	20	04	08	23	490	510	1000

L–Lecture|T–Tutorial|P–Practical|C–Credit|CA–ContinuousAssessment|ES–EndSemester

Module V	SPECTRAL DENSITIES AND LINEAR SYSTEMS WITH RANDOM INPUTS	12 Hours
Auto correlation-cross correlation, power spectral density ,cross spectral density, Properties-Wiener-Khintchine relation- relationship between cross power spectrum and correlation function. Linear time invariant system- system transfer function, Linear system with random inputs, White noise.		
		Total: 45+15 Hours
Further Reading:		
Probabilistic manner which evolve with time		
Discrete time Markov chains in modeling Electronic systems.		
Course Outcomes:		
After completion of the course, Student will be able to:		
CO1: Apply probability techniques to analyze the performance of electronic systems.		
CO2: Apply standard probability distributions to describe real-life phenomena.		
CO3: Solve problems involving multiple random variables using marginal and conditional distributions.		
CO4: Apply probability techniques such as Markov processes to analyze time-evolving systems.		
CO5: Interpret the response of random inputs to linear time-invariant systems.		
Text Books:		
- O.C.Ibe,Fundamentals of Applied Probability and Random Processes,Elsevier,1st Indian Reprint, 2007 - H.Pishro-Nik,"Introductiontoprobability,statistics,andrandomprocesses",2014.		
References:		
- D.Gross and C.M.Harris, Probability and random processes, Wiley Student edition,2004. - Yates.R.D. and Goodman. D.J.,“ Probability and Stochastic Processes”,2nd Edition,Wiley India Pvt.Ltd., Bangalore,2012. - Stark.H.,and Woods .J.W.,“ Probability and Random Processes with Applications to Signal Processing”, 3rdEdition,Pearson Education, Asia, 2002. - Miller.S.L.and Childers.D.G.,“Probability and Random Processes with Applications to Signal Processing And Communications”,Academic Press,2004.		

1902BM401		BIO MECHANICS									L	T	P	C
											3	0	0	3
		(For B.E,BME)												
Course Objectives:		The student should be made to:												
		To explain the principle of mechanics.												
		Discuss the mechanics of physiological systems.												
		Explain the mechanics of joints.												
		Illustrate the mathematical models used in the analysis of biomechanical systems.												
		To understand about Kinetics and kinematics.												
COs Vs POs MAPPING:														
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		
CO1	3	2	2	1	2	-	-	-	-	-	-	1		
CO2	3	2	2	1	-	-	-	-	-	-	-	1		
CO3	3	2	2	-	-	-	-	-	-	-	-	1		
CO4	3	2	-	-	-	-	-	-	-	-	-	1		
CO5	3	2	-	-	-	-	-	-	-	-	-	1		
AVG	3	2	2	1	2	-	-	-	-	-	-	1		
COs Vs PSOs MAPPING:														
				COs	PSO1	PSO2	PSO3							
				CO1	-	3	2							
				CO2	-	3	2							
				CO3	-	-	-							
				CO4	-	-	-							
				CO5	-	-	-							
				AVG	-	3	2							
UNIT I	INTRODUCTION TO MECHANICS											9 Hours		
Introduction – Scalars and vectors, Statics – Force types, Resolution and composition of forces, Moments of force and couple, Resultant force determination, parallel forces in space, equilibrium of coplanar forces, Dynamics, Basic principles – Linear motion, Newton’s laws of motion, Impulse and Momentum, Work and Energy Kinetics – Velocity and acceleration, Kinematics – Link segment models, Force transducers, Force plates, Introduction to Constitutive equations – Constitutive equations of Non viscous fluid, Newtonian Viscous fluid and Hookean Elastic solid														
UNIT II	BIOFLUID MECHANICS											9 Hours		
Intrinsic fluid properties – Density, Viscosity, Compressibility and Surface Tension, Viscometers – Capillary, Coaxial cylinder and cone and plate, Rheological properties of blood, Pressure-flow relationship for Non-Newtonian Fluids, Fluid mechanics in straight tube – Steady Laminar flow, Turbulent flow, Flow development, Viscous and Turbulent Shear Stress, Effect of pulsatility, Boundary Layer Separation, Structure of blood vessels, Material properties and modeling of Blood vessels, Heart –Cardiac muscle characterisation, Native heart valves – Mechanical properties and valve dynamics, Prosthetic heart valve fluid dynamics.														
UNIT III	BIO SOLID MECHANICS											9 Hours		

Constitutive equation of viscoelasticity – Maxwell & Voight models, anisotropy, Hard Tissues – Structure, blood circulation, elasticity and strength, viscoelastic properties, functional adaptation, Soft Tissues – Structure, functions, material properties and modeling of Soft Tissues–Cartilage,TendonsandLigamentsSkeletalMuscle–Muscleaction, Hil’s models, mathematical modeling, Bone fracture mechanics, Implants for bone fractures.		
UNITIV	BIOMECHANICSOFJOINTS	9 Hours
Skeletaljoints,forcesandstressesinhumanjoints,Analysisofrigidbodiesinequilibrium,Freebodydiagrams, Structureofjoints, Typesofjoints,Biomechanicalanalysisofelbow,shoulder,spinalcolumn,hip,kneeandankle, Lubrication of synovial joints, Gait analysis, Motion analysis using video.		
UNITV	MODELINGANDERGONOMICS	9 Hours
Introduction to Finite Element Analysis, finite element analysis of lumbar spine; Ergonomics–Musculoskeletal disorders, Ergonomic principles contributing to good workplace design, Design of a Computer workstation, Whole body vibrations, Hand transmitted vibrations.		
		Total: 45 Hours
Further Readings:		
Basic orthopaedic biomechanics & mechano-biology, Mechanical tribology: materials, characterization, and applications		
Course Outcomes:		
	After completion of the course, Student will be able to	
	CO1: Understand the basic principles of mechanics.	
	CO2: Outline the properties of Intrinsic fluid and principles of bio fluid dynamics.	
	CO3: Explain the fundamentals of bio-solid mechanics and implants.	
	CO4: Apply the knowledge of joint mechanics and analyze the motion of joints.	
	CO5: Analyze the spine using finite element analysis and ergonomics .	
Text Books:		
1. Y.C.Fung,—Bio-Mechanics-Mechanical Properties of Tissues ,Springer-Verlag,1998.		
2. Subrata Pal,—Text book of Bio mechanics ,Viva Books PrivateLimited,2009.		
References:		
1. Krishna B.Chandran, Ajit P. Yoganathan and Stanley E. Rittgers,—Bio fluid Mechanics: The Human Circulation , Taylor and Francis,2007.		
2. Sheraz S. Malik and Shahbaz S.Malik,—Orthopaedic Biomechanics Made Easy , Cambridge University Press, 2015		
3. Jay D. Humphrey, Sherry De Lange,—An Introduction to Biomechanics: Solids and Fluids, Analysis and Design , Springer Science Business Media,2004.		

1902BM402		BASICS OF PATHOLOGY AND MICROBIOLOGY									L	T	P	C																											
											3	0	0	3																											
Course Objectives:																																									
1. To understand the structural and functional aspects of living organisms.																																									
2. To know the etiology and remedy in treating the pathological diseases.																																									
3. To practice on chemical and structural examinations.																																									
4. To know about the microbial cultures.																																									
5. To learn about the antibody reactions and diseases caused by microbes.																																									
COs Vs POs MAPPING:																																									
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12																													
CO1	3	3	1	-	-	-	-	-	-	-	-	-	2																												
CO2	3	2	1	-	-	-	-	-	-	-	-	-	2																												
CO3	3	2	1	-	-	-	-	-	-	-	-	-	-																												
CO4	3	2	1	-	-	-	-	-	-	-	-	-	-																												
CO5	3	2	1	-	-	-	-	-	-	-	-	-	-																												
AVG	3	2.2	1	-	-	-	-	-	-	-	-	-	2																												
COs Vs PSOs MAPPING:																																									
<table><tr><td>COs</td><td>PSO1</td><td>PSO2</td><td>PSO3</td></tr><tr><td>CO1</td><td>2</td><td>-</td><td>-</td></tr><tr><td>CO2</td><td>2</td><td>-</td><td>-</td></tr><tr><td>CO3</td><td>-</td><td>-</td><td>-</td></tr><tr><td>CO4</td><td>-</td><td>-</td><td>-</td></tr><tr><td>CO5</td><td>-</td><td>-</td><td>-</td></tr><tr><td>AVG</td><td>2</td><td>-</td><td>-</td></tr></table>														COs	PSO1	PSO2	PSO3	CO1	2	-	-	CO2	2	-	-	CO3	-	-	-	CO4	-	-	-	CO5	-	-	-	AVG	2	-	-
COs	PSO1	PSO2	PSO3																																						
CO1	2	-	-																																						
CO2	2	-	-																																						
CO3	-	-	-																																						
CO4	-	-	-																																						
CO5	-	-	-																																						
AVG	2	-	-																																						
UNIT I	CELL DEGENERATION, REPAIR AND NEOPLASIA											9 Hours																													
Cell injury and Necrosis-Apoptosis, Intra cellular accumulations, Pathological calcification – cellular adaptations of growth and differentiation, Inflammation and Repair including fracture healing, Neoplasia -Classification, Benign and Malignant tumours, carcino genesis, spread of tumours. Autopsy and biopsy.																																									
UNIT II	FLUID AND HEMO DYNAMIC DERRANGEMENTS											9 Hours																													
Edema – normal hemostasis - thrombosis, disseminated intra vascular coagulation, embolism, infarction, shock. Hematological disorders – Bleeding disorders, Leukaemias, Lymphomas.																																									
UNIT III	MICROSCOPES											9 Hours																													
Light microscope – brightfield, dark field, phase contrast, fluorescence, Electron microscope (TEM & SEM). Preparation of samples for electron microscope. Staining methods – simple, gram staining and AFB staining. Parts of compound microscope - Flow cytometry and its applications.																																									
UNIT IV	MICROBIAL CULTURES											9 Hours																													
Morphological features and structural organization of bacteria – growth curve, Sterilization techniques – physical and Chemical methods, identification of bacteria, culture media and its types, culture techniques and observation of culture.																																									
UNIT V	IMMUNOLOGY											9 Hours																													

Natural and artificial immunity- opsonization- phagocytosis, inflammation, Immune deficiency syndrome, antibodies and its types, antigen and antibody reactors, immunological techniques – immune diffusion, immune electrophoresis, radio immune assay and enzyme linked immune sorbent assay, monoclonal antibodies. Disease caused by bacteria, fungi, protozoal, virus and helminths.	
	Total: 45 Hours
Further Reading: • Comprehend and appreciate the significance and role of this course in the present contemporary world. • Analyze structural and functional aspects of living organisms. • Explain the function of microscopes. • Discuss on the importance of public health. • Describe treatment methods involved incurring the pathological diseases.	
After completion of the course, Student will be able to	
CO1: Analyze the Structural and functional aspects of living organisms.	
CO2:Understand the fluid and hemodynamic disorders.	
CO3:Explain the function, types and parts of Microscopes and their applications.	
CO4:Describe methods involved in microbial cultures.	
CO5:Understand the natural and artificial immunity and different immunological techniques.	
Text Books:	
1. Harsh Mohan, “Text book of Pathology”. Jaypee Brothers Medical publishers private Limited, 7th Edition, 2014. 2. Anantha Narayanan ,“Text Book of Microbiology”,OrientLongman,6thedition,2012.	
References:	
1. Ramzi S Cotran, Vinay Kumar & Stanley L Robbins, “Pathologic Basis of diseases”, WB Saunders Co.7th Edition, 2005. 2. Underwood JCE,“General and Systematic Pathology”, Churchill Living stone,3rd,Edition,2000. 3. Anantha narayanan,“Microbiology”,Panicker University press.9thEdition,2013 4. Prescott,Harley,Klein,“Microbiology”,Mc Graw Hill,9th Edition,2013.	

1902BM403		CONTROL SYSTEMS	L	T	P	C
			3	0	0	3

Course Objectives:

1. This course aims to introduce students to the principles and applications of control systems.
2. It covers basic concepts such as block diagram reduction and time-domain analysis solutions for time-invariant systems.
4. The course also deals with various aspects of stability analysis in both frequency and time domains.
5. Students will develop an understanding of the practical applications of control systems.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	-	2	2	-	-	-	-	2
CO2	3	3	-	-	-	2	-	-	-	-	-	2
CO3	3	3	-	-	-	-	-	-	-	-	-	2
CO4	3	3	-	-	-	-	-	-	-	-	-	2
CO5	3	3	-	-	-	-	-	-	-	-	-	-
AVG	3	3	-	-	-	2	2	-	-	-	-	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	-	-
CO2	-	-	-
CO3	-	-	-
CO4	-	-	-
CO5	-	-	-
AVG	-	-	-

Module I	INTRODUCTION OF CONTROL SYSTEMS	9 Hours
Basic concept of control systems - Open loop, closed loop control systems and their differences, Block diagram algebra, Representation by signal flow graph-Reduction using Mason's gain formula-Feedback characteristics and effect of Feedback		
Module II	TIME RESPONSE ANALYSIS	9 Hours
Time response analysis-Time response of first order system, Transient response of second order system, Time domain specification-steady state response, Steady state error, Effect of proportional, derivatives, Proportional integral system		
Module III	FREQUENCY RESPONSE ANALYSIS	9 Hours
Frequency response -Frequency domain specification, stability analysis from bode plot, polar plot, Nyquist plot- Compensation techniques-Lag, Lead, lead-lag controllers design in frequency domain.		
Module IV	STABILITY ANALYSIS AND ROOT LOCUS TECHNIQUES	9 Hours
Concept of stability-Routh Hurwitz criterion, Nyquist stability criterion-Routh locus concept-construction of root locus		
Module V	APPLICATIONS OF CONTROL SYSTEMS	9 Hours

Aircraft flight control systems-Director (military),Embedded instrumentation-Fire control system-Guidance Navigation and control-Laser ignition-Weight shift control		
	Total:	45 Hours
Further Reading:		
Bio control and disease modeling		
Course Outcomes:		
After completion of the course, Student will be able to		
CO1:Have a better understanding on open loop and closed loop control system, concept of feedback in control systems.		
CO2: apply Transfer function representation through block diagram algebra and signal flow graph, time response analysis.		
CO3:Frequency response analysis through bode plot polar plot, Nyquist plot and basics of state space analysis.		
CO4:Better understanding on stability analysis and root locus techniques.		
CO5:Apply control system concepts on real time applications.		
Text Books:		
- I.J.Nagrath, Madan Gopal “ Text book of control system Engineering” by, New Age International, 2008 - B.S.Manke,“Control system design”, Stylus publishing,2017.		
References:		
1. Automatic control systems, third edition, Benjamin C.Kuo.		
2. Control and Dynamical Systems, Karl Johan Aström° Richard M. Murray, Versionv2.10c(March 4,2010),prince on University press.		
3. Modern Control Systems, twelfth edition ,Richard C. Dorf University of California, Davis,Robert H.Bishop Marquette University.		
4. Katshui koogata,“Modern Control engineering”,5 th edition,2012		

1902BM404		BIOMEDICAL INSTRUMENTATION						L	T	P	C																													
								3	0	0	3																													
Course Objectives:																																								
1. To understand the basic theory of Biopotential Electrode sand Biopotential measurement.																																								
2. To Understand the design of Biopotential amplifiers.																																								
3. To know about bioelectric signals and amplifiers																																								
4. To analyse the biomedical recording systems																																								
5. Study the various non-electrical physiological measurement and biochemical measurements.																																								
COs Vs POs MAPPING:																																								
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12																												
CO1	3	3	-	-	-	3	2	-	-	-	-	2																												
CO2	3	3	-	-	-	3	2	-	-	-	-	2																												
CO3	3	3	-	-	-	3	-	-	-	-	-	2																												
CO4	3	3	-	-	-	-	-	-	-	-	-	2																												
CO5	3	3	-	-	-	-	-	-	-	-	-	2																												
AVG	3	3	-	-	-	3	2	-	-	-	-	2																												
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COs	PSO1	PSO2	PSO3																																					
CO1	2	2	-																																					
CO2	2	2	-																																					
CO3	2	-	-																																					
CO4	2	-	-																																					
CO5	2	-	-																																					
AVG	2	2	-																																					
UNIT I	BIOPOTENTIAL ELECTRODES										9 Hours																													
Origin of bio potential and its propagation. Electrode-electrolyte interface, electrode–skin interface, half-cell potential, Contact impedance, polarization effects of electrode–non polarizable electrodes. Types of electrodes - surface, needle and micro electrodes and their equivalent circuits. Recording problems-motion artifacts, measurement with two electrodes.																																								
UNIT II	BIOPOTENTIAL MEASUREMENTS										9 Hours																													
Bio signals characteristics–frequency and amplitude ranges. ECG–Einthovens triangle, standard12 lead system, Principles of vector cardiography. EEG –10-20 electrode system, unipolar, bipolar and average mode. EMG – unipolar and Bipolar mode. Recording of ERG, EOG and EGG																																								
UNIT III	BIOELECTRIC AMPLIFIERS										9 Hours																													
Need for bio-amplifier-single ended bio-amplifier, differential bio-amplifier, Impedance matching circuit, isolation amplifiers–transformer and optical isolation-isolated DC amplifier and AC carrier amplifier., Power line interference, Right leg driven ECG amplifier, Band pass filtering																																								
UNIT IV	MEASUREMENT OF NON-ELECTRICAL PARAMETERS										9 Hours																													
Temperature, respiration rate and pulse rate measurements. Blood Pressure: indirect methods -Auscultatory method, oscillometric method, direct methods: electronic manometer, Pressure amplifiers, Systolic, diastolic, mean detector circuit. Blood flow and cardiac output measurement: Indicator dilution, thermal dilution and dye dilution method, Electromagnetic and ultrasound blood flow measurement.																																								
UNIT V	PATIENT MONITORING SYSTEMS										9 Hours																													

System concepts-Cardiac monitor-selection of system parameters, Bedside monitors, Central monitors, Heart rate meter, Pulserate meter, sphygmo manometers - Holter monitor and Cardiac stress test, Cardiac cauterization instrumentation-Organization and equipment used in ICCU & ITU.		
	Total:	45 Hours
Further Reading:		
• Medical Imaging, Biomedical Image Processing.		
Course Outcomes:		
After completion of the course, Student will be able to		
CO1:Understand the basics of bio potential electrodes and its types.		
CO2:Describe the bio signal characteristics and fundamentals of Biopotential recording.		
CO3:Design and analyze the various bio amplifiers.		
CO4:Measure various physiological and biochemical parameters.		
CO5:Understand the concepts of patient monitoring systems and equipment's in ICUs		
Text Books:		
1. Joseph J. Carr and John M. Brown, Introduction to Biomedical Equipment Technology, 4th Edition, Pearson Education, 2014.		
2. R.S. Khandpur, Handbook of Biomedical Instrumentation, 2014.		
References:		
1. John G. Webster, <i>Medical Instrumentation: Application and Design</i> , 4th Edition, John Wiley and Sons, New York, 2009.		
2. R.S. Khandpur, <i>Handbook of Biomedical Instrumentation</i> , 3rd Edition, Tata McGraw Hill, New Delhi, 2014.		
3. L.A. Geddes and L.E. Baker, <i>Principles of Applied Biomedical Instrumentation</i> , 3rd Edition, Reprint 2008, John Wiley and Sons.		
4. Leslie Cromwell, Fred J. Weibell, Erich A. Pfeiffer, <i>Biomedical Instrumentation and Measurements</i> , 2nd Edition, Pearson Education India, 2015.		
5. Myer Kutz, <i>Standard Handbook of Biomedical Engineering & Design</i> , McGraw-Hill, 2003.		

1902BM405		DIGITAL ELECTRONICS AND INTEGRATED CIRCUITS								L	T	P	C																												
										3	0	0	3																												
Course Objectives:																																									
1. To study the circuit configurations and introduce practical applications of linear integrated circuits.																																									
2. To introduce the concepts and applications of ADC (Analog-to-Digital Converter) and DAC (Digital-to-Analog Converter) in real-time systems, as well as the Phase Locked Loop (PLL) and its applications.																																									
3. To introduce number systems and logic gates.																																									
4. To analyze combinational logic circuits.																																									
5. To explore the analysis and design procedures for sequential circuits.																																									
COs Vs POs MAPPING:																																									
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12																													
CO1	3	3	3	3	-	-	-	-	3	3	2	2																													
CO2	3	3	3	3	-	-	-	-	3	3	2	2																													
CO3	3	3	3	3	-	-	-	-	-	-	-	-																													
CO4	3	3	3	3	-	-	-	-	3	3	2	2																													
CO5	3	3	3	3	-	-	-	-	-	-	-	-																													
AVG	3	3	3	3	-	-	-	-	3	3	2	2																													
COs Vs PSOs MAPPING:																																									
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COs	PSO1	PSO2	PSO3																																						
CO1	3	-	-																																						
CO2	3	-	-																																						
CO3	-	-	-																																						
CO4	-	-	-																																						
CO5	-	-	-																																						
AVG	3	-	-																																						
UNIT I	INTRODUCTION TO OPERATIONAL AMPLIFIER AND ITS APPLICATIONS										9 Hours																														
Operational amplifier –ideal characteristics, Performance Parameters, Linear and Nonlinear Circuits and their analysis- voltage follower, Inverting amplifier, Noninverting Amplifiers, Differentiator, Integrator, Voltage to Current converter, Instrumentation amplifier-Low pass, High pass filter and band pass filters, Comparator – Multivibrator and Schmitt trigger- Triangular wave generator.																																									
UNIT II	DIGITAL TO ANALOG AND ANALOG TO DIGITAL CONVERTERS										9 Hours																														

Analog switches- High speed sample and hold circuit and IC,,s,-Types of D/A converter -Weighted resistor, R-2R ladder DAC, D/A Accuracy and Resolution. A/D converter - Flash, Dual slope, Successive approximation, A/D Accuracy and Resolution. Voltage controlled oscillator – Voltage to Frequency converters.		
UNIT III	THEBASICGATESANDCOMBINATIONALALLOGICCIRCUITS	9 Hours
Number Systems – Decimal, Binary, Octal, Hexadecimal, 1’s and 2’s complements, Codes – Binary, BCD, 8421, 2421, Excess3, Biquinary, Gray, Alphanumeric codes, Boolean theorems, Logic gates, Universal gates, Sum of products and Product of sums, Minterms and Maxterms, Karnaugh map and Tabulation methods.		
UNIT IV	COMBINATIONAL LOGIC CIRCUITS	9 Hours
Problem formulation and design of combinational circuits – Code - Converters, Half and Full Adders, Binary Parallel Adder – Carry look ahead Adder, BCD Adder, Magnitude Comparator, Decoder, Encoder, Priority Encoder, Mux/Demux, Implementation of combinational logic using standard ICs-ROM, PLA and PAL.		
UNIT V	SEQUENTIAL LOGIC CIRCUITS	9 Hours
Flipflops – SR, JK, T, D, Master/Slave FF, Triggering of FF, Analysis and design of clocked sequential circuits – state minimization, state assignment, circuit implementation. Counters, Ripple Counters, Ring Counters.		
		Total: 45 Hours
Further Reading:		
• Study about Timer IC 555, Monolithic PLL IC 565, IC 723 general purpose regulator		
Course Outcomes:		
After completion of the course, Student will be able to		
CO1: Ability to design new analog linear circuits and develop linear IC-based systems.		
CO2: Understand the concept and application of ADC and DAC in real-time systems, as well as Phase Locked Loop (PLL) and its applications.		
CO3: Use Boolean algebra and apply it to digital systems.		
CO4: Design various combinational digital circuits using logic gates.		
CO5: Perform analysis and design of synchronous and asynchronous sequential circuits.		
Text Books:		
1. Choudhary D. Roy, Linear Integrated Circuits, New Age International Publishers, 2018.		
2. Ken Martin, Digital Integrated Circuit Design, Oxford University Press, 2012.		
References:		
1. Sergio Franco, Design with Operational Amplifiers and Analog Integrated Circuits, 3rd Edition, McGraw Hill Education, 2017.		
2. M. Morris Mano and Michael D. Ciletti, Digital Design, 5th Edition, Pearson, 2013.		
3. Charles H. Roth, Jr., Fundamentals of Logic Design, 7th Edition, Jaico Books, 2013.		
4. Gray and Meyer, Analysis and Design of Analog Integrated Circuits, Wiley International, 2009.		

1901MCX01		ENVIRONMENTAL SCIENCE AND ENGINEERING	L	T	P	C
			2	0	0	0
		(For B.E., BME)				

Course Objectives:	The student should be made to:
	1. To study the nature and facts about the environment.
	2. To find and implement scientific, technological, economic, and political solutions to environmental problems.
	3. To study the interrelationship between living organisms and the environment.
	4. To appreciate the importance of the environment by assessing its impact on the human world; to envision the surrounding environment, its functions, and its value.
	5. To study the dynamic processes and understand the features of the Earth's interior and surface.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	2	-	-	-	2	2	2	2
CO2	3	2	2	2	-	-	-	-	2	2	2	2
CO3	3	2	2	2	-	-	-	-	2	-	2	2
CO4	3	2	2	-	-	-	-	-	2	-	-	2
CO5	3			-	-	-	-	-	2	-	-	2
AVG	3	2.6	2.2	2.4	2	-	-	-	2	2	2	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	3	3	3
CO2	2	3	-
CO3	2	-	-
CO4	2	-	-
CO5	2	-	-
AVG	2.4	3	3

UNIT I	ENVIRONMENT, ECOSYSTEMS AND BIODIVERSITY	9 Hours
Definition, scope and importance of environment – need for public awareness - concept of an ecosystem – structure and function of an ecosystem – producers, consumers and decomposers – energy flow in the ecosystem – ecological succession – foodchains, food webs and ecological pyramids – Introduction, types, characteristic features, structure and function of the (a) forest ecosystem (b) grassland ecosystem (c) desert ecosystem (d) aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries) – Introduction to biodiversity definition: genetic, species and ecosystem diversity – bio geographical classification of India – value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values – Biodiversity at global, national and local levels– India as a mega-diversity nation– hot-spots of biodiversity – threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – endangered and endemic species of India –conservation of biodiversity: In-situandex-situ conservation of biodiversity. Field study of common plants, insects, birds; Field study of simple ecosystems–pond, river, hillslopes, etc		

UNIT II	ENVIRONMENTAL POLLUTION	9 Hours
Definition – causes, effects and control measures of: (a) Air pollution (b) Water pollution (c) Soil pollution (d) Marine pollution (e) Noise pollution (f) Thermal pollution (g) Nuclear hazards – solid waste management: causes, effects and control measures of municipal solid wastes – role of an individual in prevention of pollution – pollution case studies – disaster management: floods, earthquake, cyclone and landslides. Field study of local polluted site – Urban/Rural/Industrial/Agricultural.		
UNIT III	NATURAL RE SOURCES	9 Hours
Forest resources: Use and over-exploitation, deforestation, case studies- timber extraction, mining, dams and their effects on forests and tribal people – Water resources: Use and over- utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems – Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies – Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies – Energy resources: Growing energy needs, renewable and non renewable energy sources, use of alternate energy sources. case studies – Land resources: Land as a resource, land degradation, man induced landslides, soil erosion and desertification – role of an individual in conservation of natural resources – Equitable use of resources for sustainable lifestyles. Field study of local area to document environmental assets – river/forest/grassland/hill/mountain.		
UNIT IV	SOCIAL ISSUES AND THE ENVIRONMENT	9 Hours
From unsustainable to sustainable development – urban problems related to energy – water conservation, rain water harvesting, watershed management – resettlement and rehabilitation of people; its problems and concerns, case studies – role of non-governmental organization- environmental ethics: Issues and possible solutions – climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust, case studies. – wasteland reclamation – consumerism and waste products – environment production act – Air (Prevention and Control of Pollution) act – Water (Prevention and control of Pollution) act – Wild life protection act – Forest conservation act – enforcement machinery Involved in environmental legislation – central and state pollution control boards – Public awareness.		
UNIT V	HUMAN POPULATION AND THE ENVIRONMENT	9 Hours
Population growth, variation among nations – population explosion – family welfare programme – environment and Human health – human rights – value education – HIV/AIDS – women and child welfare – role of information technology in environment and human health – Case studies		
		Total: 45 Hours
Further Reading:	Analyze the continuous pollution signals & systems and its bio signal applications.	
Course Outcomes:		
	After completion of the course, Student will be able to	
	CO1: Understand the environmental issues, drawing from ecological, social, physical, economic, and legal perspectives	
	CO2: Gain expertise in identifying, assessing, and mitigating various forms of pollution, including air, water, and soil pollution	
	CO3: Understand the principles of managing resources like water, forests, minerals, and land, emphasizing sustainable practices.	
	CO4: Understand the Environmental degradation, like pollution and climate change, which often disproportionately affects vulnerable populations, exacerbating existing social inequalities	
	CO5: Understand the population growth, human rights, welfare and human health.	
Text Book:		
1. Benny Joseph, <i>Environmental Science and Engineering</i> , Tata McGraw-Hill, New Delhi, 2006.		
2. Gilbert M. Masters, <i>Introduction to Environmental Engineering and Science</i> , 2nd Edition, Pearson Education, 2004.		
References:		
1. Benny Joseph, <i>Environmental Science and Engineering</i> , Tata McGraw-Hill, New Delhi, 2006.		
2. Dharmendra S. Sengar, <i>Environmental Law</i> , Prentice Hall of India Pvt Ltd, New Delhi, 2007.		
3. Erach Bharucha, <i>Textbook of Environmental Studies</i> , Universities Press (India) Pvt Ltd, Hyderabad, 2015.		

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|---|
| 4. R. Rajagopalan, <i>Environmental Studies - From Crisis to Cure</i> , Oxford University Press, 2005. |
| 5. G. Tyler Miller and Scott E. Spoolman, <i>Environmental Science</i> , Cengage Learning India Pvt Ltd, Delhi, 2014. |

1902BM451		PATHOLOGY AND MICROBIOLOGY LABORATORY	L	T	P	C
			0	0	2	1

Course Objectives:

1. Students should learn how to use a compound microscope and practice chemical examinations.
2. To learn how to use cryoprocessing and perform histopathological examinations.
3. To understand the concepts of staining techniques.
4. To perform microscopic visualization of microorganisms for disease determination.
5. To gain practical knowledge of techniques used in tissue processing.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3	-	-	-	-	-	2	2
CO2	3	3	3	3	-	-	-	-	-	-	2	2
CO3	3	3	3	3	3	-	-	-	-	-	-	-
CO4	3	3	3	2	-	-	-	-	-	-	-	-
CO5	3	3	2	2	-	-	-	-	-	-	-	-
AVG	3	3	2.8	2.4	3	-	-	-	-	-	2	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2	PSO3
CO1	-	3	-
CO2	-	-	-
CO3	-	-	-
CO4	-	-	-
CO5	-	-	-
AVG	-	3	-

List of Experiments:

1. Study of bone marrow charts.
2. Steam sterilization using autoclave.
3. Manual paraffin tissue processing and section cutting.
4. Cryoprocessing of tissue and cryosectioning.
5. Basic staining techniques – Hematoxylin and Eosin staining.
6. Special stains – Cresyl Fast Blue (CFV), Trichrome, Oil Red O, PAS (Periodic Acid-Schiff).
7. Simple stain, Gram stain, Acid-Fast Bacilli (AFB) stain.
8. Microscopic examination of slides showing malarial parasites, microfilaria, and *Leishmania donovani*.
9. Hematology slides of anemia and leukemia.
10. Physical and chemical examination of urine (testing for protein, reducing substances, ketones, bilirubin, and blood).

Additional Experiments:

1. Study of histopathological slides of benign and malignant tumors.
2. Differential count of various white blood cells (WBCs) and blood group identification.

Course Outcomes:

After completion of the course, Student will be able to

CO1:List the laboratory tests performed for urine analysis.

CO2:Staining procedures are explained completely for the identification of microorganisms.
CO3:Practical knowledge about microscopes.
CO4:Explains the bleeding and clotting time for analysis of the fluid in the body.
CO5:Practical description about the structure and organization of microorganisms.
Text book:
1. Harsh Mohan, <i>Textbook of Pathology</i> , 7th Edition, Jaypee Brothers Medical Publishers Private Limited, 2014.
References:
1. William B. Coleman and Gregory J. Tsongalis, <i>Molecular Pathology: The Molecular Basis of Human Disease</i> .
2. Ramzi S. Cotran, Vinay Kumar, and Stanley L. Robbins, <i>Pathologic Basis of Diseases</i> , 7th Edition, W.B. Saunders Co., 2005.
3. J.C.E. Underwood, <i>General and Systematic Pathology</i> , 3rd Edition, Churchill Livingstone, 2000.

1902BM452		BIOMEDICAL INSTRUMENTATION LABORATORY								L	T	P	C																												
										0	0	2	1																												
Course Objectives:																																									
1. To study and design bioamplifiers.																																									
2. To provide hands-on training on the measurement of physiological parameters.																																									
3. To study blood flow and blood measurement techniques.																																									
4. To understand patient monitoring systems.																																									
5. To study the working and applications of pH meters.																																									
COs Vs POs MAPPING:																																									
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12																													
CO1	3	2	1	-	-	3	-	-	-	-	2	3																													
CO2	3	2	1	-	-	-	-	-	-	-	2	3																													
CO3	3	2	1	-	3	-	-	-	-	-	-	3																													
CO4	3	2	1	-	-	-	-	-	-	-	-																														
CO5	3	2	1	-	-	-	-	-	-	-	-																														
AVG	3	2	1	-	3	3	-	-	-	-	2	3																													
COs Vs PSOs MAPPING:																																									
<table><tr><td>COs</td><td>PSO1</td><td>PSO2</td><td>PSO3</td></tr><tr><td>CO1</td><td>-</td><td>-</td><td>-</td></tr><tr><td>CO2</td><td>-</td><td>-</td><td>-</td></tr><tr><td>CO3</td><td>-</td><td>-</td><td>-</td></tr><tr><td>CO4</td><td>-</td><td>-</td><td>-</td></tr><tr><td>CO5</td><td>-</td><td>-</td><td>-</td></tr><tr><td>AVG</td><td>-</td><td>-</td><td>-</td></tr></table>														COs	PSO1	PSO2	PSO3	CO1	-	-	-	CO2	-	-	-	CO3	-	-	-	CO4	-	-	-	CO5	-	-	-	AVG	-	-	-
COs	PSO1	PSO2	PSO3																																						
CO1	-	-	-																																						
CO2	-	-	-																																						
CO3	-	-	-																																						
CO4	-	-	-																																						
CO5	-	-	-																																						
AVG	-	-	-																																						
List of Experiments:																																									
1. Design of low noise pre-amplifier.																																									
2. Recording of ECG signal and analysis using simulation.																																									
3. Simulation and analysis of EMG amplifier.																																									
4. Simulation and analysis of EEG using simulation tools.																																									
5. Measurement of respiration rate.																																									
6. Measurement of blood flow velocity using ultrasound transducer.																																									
7. Measurement of blood pressure using sphygmomanometer.																																									
8. Study of characteristics of baby ventilator.																																									
9. Measurement of vital parameters using patient monitoring system.																																									
10. Measurement of hydrogen ion concentration using pH meter.																																									
Additional Experiments:																																									
1. Understand and implement isolation techniques in designing biomedical instruments.																																									
2. Measure vital parameters using patient monitoring systems.																																									
Course Outcomes:																																									
After completion of the course, Student will be able to																																									
CO1: Design the amplifier for Bio signal measurements .																																									

CO2: Measure heart rate and heart sounds.
CO3: Record and analyze pulse rate and respiration rate.
CO4: Measure blood pressure and blood flow.
CO5:Design isolation amplifier.
Text Book:
1. Richard Aston, <i>Principles of Biomedical Instrumentation and Measurement</i> .
References:
1. John G. Webster, <i>Medical Instrumentation – Application and Design</i> .
2. Richard S.C. Cobbold, <i>Transducers for Biomedical Measurements: Principles and Applications</i> .
3. Ernest O. Doebelin, <i>Measurement Systems, Application and Design</i> .

CO3:Design Sequential Circuits using logic gates
C04:Design wave form generators and analyse their characteristics
CO5:Simulate and analyze circuits using ICs
Text Book:
1. M. Morris Mano, <i>Digital Design</i> , 4th Edition, Prentice Hall of India Pvt. Ltd., 2008 / Pearson Education (Singapore) Pvt. Ltd., New Delhi, 2003.
References:
1. Ronald J. Tocci, Neal S. Widmer, and Gregory L. Moss, <i>Digital Systems: Principles and Applications</i> , 10th Edition, Pearson Prentice Hall, 2007.
2. Joseph Cavanagh, <i>Verilog HDL: Digital Design and Modeling</i> , Taylor & Francis, 2007.
3. John F. Wakerly, <i>Digital Design</i> , 4th Edition, Pearson / PHI, 2008.